

Diag Solutions

1. 1. A sum of Rs. 1600 is to be used to give 5 cash prizes to students of a school for their overall academic performance. If each prize is Rs. 20 less than its preceding prize, find the value of each of the prizes.

Answer:

Rs. 400, Rs. 380, Rs. 360, Rs. 340, Rs. 320

Solution:

1. Let the prizes be $a, a - 20, a - 40, a - 60, a - 80$.
2. Use the sum formula $S_n = \frac{n}{2}[2a + (n - 1)d]$.
3. Substitute $n = 5$, $S_5 = 1600$, and $d = -20$.
4. Solve for a .
5. List all 5 terms.

Derivation:

$$\begin{aligned}1600 &= \frac{5}{2}[2a + (5 - 1)(-20)] \\3200 &= 5[2a - 80] \\640 &= 2a - 80 \\2a &= 720 \\ \implies a &= 360 \text{ (Wait, check: } 5a + 10d = 1600/2 \cdot 2 \text{ is wrong)} \\1600 &= \frac{5}{2}(2a - 80) \\ \implies 640 &= 2a - 80 \\ \implies 2a &= 720 \\ \implies a &= 360\end{aligned}$$

Arithmetic Progressions — Application of AP

2. 2. Find the sum of all three-digit numbers which leave a remainder of 3 when divided by 5.

Answer:

98550

Solution:

1. Identify the first three-digit number a such that $a \equiv 3 \pmod{5}$, which is 103.
2. Identify the last three-digit number l such that $l \equiv 3 \pmod{5}$, which is 998.
3. Determine the number of terms n using $a_n = a + (n - 1)d$ where $d = 5$.
4. Calculate $n = \frac{l-a}{d} + 1$.

5. Calculate the sum using $S_n = \frac{n}{2}(a + l)$.

Derivation:

$$a = 103, l = 998, d = 5$$

$$998 = 103 + (n - 1)5$$

$$895 = (n - 1)5$$

$$n - 1 = 179$$

$$\implies n = 180$$

$$S_{180} = \frac{180}{2}(103 + 998) = 90(1101) = 99090$$

Arithmetic Progressions — Sum of n terms

3. **3.** Assertion (A): The common difference of the AP $5, 2, -1, -4, \dots$ is -3 . Reason (R): The common difference d of an AP $a_1, a_2, a_3, \dots$ is given by $d = a_n - a_{n-1}$.

Answer: (A)

Solution:

1. Identify terms $a_1 = 5, a_2 = 2$.
2. Calculate $d = 2 - 5 = -3$.
3. Verify Reason definition of common difference.

Derivation:

$$d = a_2 - a_1 = 2 - 5 = -3$$

Arithmetic Progressions — Common Difference

4. **4.** Find the sum of the first 20 terms of the AP: $1, 6, 11, 16, \dots$

Answer:

$$970$$

Solution:

1. Identify $a = 1, d = 5, n = 20$.
2. Use the formula $S_n = \frac{n}{2}[2a + (n - 1)d]$.
3. Substitute and calculate.

Derivation:

$$S_{20} = \frac{20}{2}[2(1) + (20 - 1)5] = 10[2 + 19(5)] = 10[2 + 95] = 10[97] = 970$$

Arithmetic Progressions — Sum of first n terms

5. **5.** How many two-digit numbers are divisible by 3?

Answer:

Solution:

1. Identify the first number $a = 12$ and last number $l = 99$.
2. Use $a_n = a + (n - 1)d$ where $d = 3$.
3. Solve for n .

Derivation:

$$\begin{aligned}
 99 &= 12 + (n - 1)3 \\
 \implies 87 &= (n - 1)3 \\
 \implies 29 &= n - 1 \\
 \implies n &= 30
 \end{aligned}$$

Arithmetic Progressions — Nth term of an AP

6. 6. If the sum of first n terms of an AP is $S_n = 3n^2 + n$, find the common difference.

Answer:

6

Solution:

1. Find $a_1 = S_1 = 3(1)^2 + 1 = 4$.
2. Find $S_2 = 3(2)^2 + 2 = 14$.
3. Find $a_2 = S_2 - S_1 = 14 - 4 = 10$.
4. Calculate $d = a_2 - a_1 = 10 - 4 = 6$.

Derivation:

$$a_1 = 4, a_2 = 10, d = 10 - 4 = 6$$

Arithmetic Progressions — Relationship between sum and terms

7. 7. The fourth term of an AP is 11 and the sum of the fifth and seventh terms is 34. Find the common difference.

Answer:

3

Solution:

1. Write $a + 3d = 11$ (eq 1).
2. Write $(a + 4d) + (a + 6d) = 34 \implies 2a + 10d = 34 \implies a + 5d = 17$ (eq 2).
3. Subtract (eq 1) from (eq 2) to find d .

Derivation:

$$\begin{aligned}
(a + 5d) - (a + 3d) &= 17 - 11 \\
\implies 2d &= 6 \\
\implies d &= 3
\end{aligned}$$

Arithmetic Progressions — System of linear equations in AP

8. **8.** Assertion (A): The common difference of an A.P. where $a_n = 3n + 7$ is 3. Reason (R): The common difference of an A.P. given by $a_n = An + B$ is always A .

Answer: (A)

Solution:

1. Find the first term by substituting $n = 1$: $a_1 = 3(1) + 7 = 10$.
2. Find the second term by substituting $n = 2$: $a_2 = 3(2) + 7 = 13$.
3. Calculate common difference $d = a_2 - a_1 = 13 - 10 = 3$.
4. Since $a_n = An + B$ implies $a_{n+1} - a_n = A(n+1) + B - (An + B) = A$, the reason is correct.

Derivation:

$$d = a_{n+1} - a_n = [3(n+1) + 7] - [3n + 7] = 3n + 3 + 7 - 3n - 7 = 3$$

Arithmetic Progressions — General term of an A.P.

9. **9.** Find the sum of all two-digit numbers which are divisible by 6.

Answer:

$$735$$

Solution:

1. Identify the first and last two-digit numbers divisible by 6: $a = 12$, $l = 96$.
2. Use the formula $a_n = a + (n - 1)d$ to find n : $96 = 12 + (n - 1)6$.
3. Solve for n : $84 = (n - 1)6 \implies n - 1 = 14 \implies n = 15$.
4. Use $S_n = \frac{n}{2}(a + l)$ to find the sum: $S_{15} = \frac{15}{2}(12 + 96)$.

Derivation:

$$n = \frac{96 - 12}{6} + 1 = 15; S_{15} = \frac{15}{2}(108) = 15 \times 54 = 735$$

Arithmetic Progressions — Sum of n terms of an A.P.

10. **10.** If the 4th term of an A.P. is 0, prove that its 25th term is triple its 11th term.

Answer:

$$a_{25} = 3a_{11}$$

Solution:

1. Express the 4th term: $a + 3d = 0 \implies a = -3d$.
2. Express the 25th term: $a_{25} = a + 24d$.
3. Substitute $a = -3d$ into a_{25} : $a_{25} = -3d + 24d = 21d$.
4. Express the 11th term: $a_{11} = a + 10d = -3d + 10d = 7d$.
5. Compare: $21d = 3(7d) \implies a_{25} = 3a_{11}$.

Derivation:

$$\begin{aligned}
 a_4 &= a + 3d = 0 \\
 &\implies a = -3d. \\
 a_{25} &= a + 24d = -3d + 24d = 21d. \\
 a_{11} &= a + 10d = -3d + 10d = 7d. \\
 21d &= 3 \times 7d.
 \end{aligned}$$

Arithmetic Progressions — General term properties

- 11. 11.** In an A.P., the sum of first n terms is $S_n = 3n^2 + 5n$. Find the common difference of the A.P.

Answer:

6

Solution:

1. Find S_1 (first term a_1): $S_1 = 3(1)^2 + 5(1) = 8$.
2. Find S_2 (sum of first two terms): $S_2 = 3(2)^2 + 5(2) = 12 + 10 = 22$.
3. Find the second term $a_2 = S_2 - S_1 = 22 - 8 = 14$.
4. Find the common difference $d = a_2 - a_1 = 14 - 8 = 6$.

Derivation:

$$a_1 = 8, a_2 = 14, d = 14 - 8 = 6$$

Arithmetic Progressions — Sum of n terms

- 12. 12.** Find the value of k for which $k - 1, k + 3, 3k - 1$ are in Arithmetic Progression.

Answer:

$k = 5$

Solution:

1. For three terms x, y, z to be in A.P., $2y = x + z$.
2. Set up the equation: $2(k + 3) = (k - 1) + (3k - 1)$.
3. Expand: $2k + 6 = 4k - 2$.
4. Solve for k : $6 + 2 = 4k - 2k \implies 8 = 2k \implies k = 4$ wait, recalculate:
 $2k + 6 = 4k - 2 \implies 8 = 2k \implies k = 4$.

Derivation:

$$2(k + 3) = k - 1 + 3k - 1$$

$$2k + 6 = 4k - 2$$

$$8 = 2k$$

$$k = 4$$

Arithmetic Progressions — Properties of A.P.

- 13. 13.** Assertion (A): The common difference of the Arithmetic Progression 5, 2, -1, -4, ... is -3. Reason (R): The common difference 'd' of an AP is given by the formula $d = a_n - a_{n-1}$.

Answer: (A)

Solution:

1. Identify the terms of the AP: $a_1 = 5$, $a_2 = 2$.
2. Calculate the common difference using $d = a_2 - a_1 = 2 - 5 = -3$.
3. Verify the Assertion: The calculated difference is -3, so Assertion (A) is true.
4. Verify the Reason: The formula $d = a_n - a_{n-1}$ is the standard definition of the common difference, so Reason (R) is true.
5. Conclusion: Since R correctly explains how to find the common difference used in A, (A) is the correct option.

Derivation:

$$d = a_2 - a_1 = 2 - 5 = -3$$

Arithmetic Progressions — Common Difference

- 14. 14.** Assertion (A): The common difference of the Arithmetic Progression 5, 2, -1, -4, ... is -3. Reason (R): The common difference d of an AP is given by $d = a_n - a_{n-1}$.

Answer: (A)

Solution:

1. Calculate the common difference using the first two terms: $d = a_2 - a_1 = 2 - 5 = -3$.
2. Verify the common difference using the next two terms: $d = a_3 - a_2 = -1 - 2 = -3$.
3. The formula for common difference $d = a_n - a_{n-1}$ is the standard definition in an AP.
4. Since both the assertion is true and the reason correctly explains the calculation of the common difference, option (A) is correct.

Derivation:

$$d = a_2 - a_1 = 2 - 5 = -3 \text{ and } d = a_3 - a_2 = -1 - 2 = -3$$

- 15. 15.** Assertion (A): The common difference of the Arithmetic Progression $5, 2, -1, -4, \dots$ is -3 . Reason (R): The common difference d of an AP is given by $a_n - a_{n-1}$.

Answer: (A)

Solution:

1. Calculate the common difference using the first two terms: $d = a_2 - a_1 = 2 - 5 = -3$.
2. Verify with the next terms: $d = a_3 - a_2 = -1 - 2 = -3$.
3. The formula for common difference is indeed $d = a_n - a_{n-1}$.
4. Since both statements are true and the reason correctly explains the assertion, the answer is (A).

Derivation:

$$d = a_2 - a_1 = 2 - 5 = -3$$

- 16. 16.** Assertion (A): The common difference of the Arithmetic Progression $5, 2, -1, -4, \dots$ is -3 . Reason (R): The common difference d of an AP is given by $d = a_n - a_{n-1}$.

Answer: (A)

Solution:

1. Identify the terms of the AP: $a_1 = 5, a_2 = 2$.
2. Calculate the common difference using the formula $d = a_2 - a_1$.
3. Substitute the values: $d = 2 - 5 = -3$.
4. Verify that the Assertion is true and the Reason correctly defines the method to find the common difference.

Derivation:

$$d = a_2 - a_1 = 2 - 5 = -3$$

- 17. 17.** Assertion (A): The common difference of the Arithmetic Progression $5, 2, -1, -4, \dots$ is -3 . Reason (R): The common difference d of an Arithmetic Progression is given by $d = a_n - a_{n-1}$.

Answer: (A)

Solution:

1. Calculate the common difference using the first two terms: $d = a_2 - a_1 = 2 - 5 = -3$.

2. Verify the result with the next terms: $d = a_3 - a_2 = -1 - 2 = -3$.
3. Confirm that the formula $d = a_n - a_{n-1}$ is the standard definition for an AP.
4. Conclude that both A and R are true and R correctly explains A.

Derivation:

$$d = a_2 - a_1 = 2 - 5 = -3$$

$\implies$ Assertion is true. Reason is the standard definition of common difference

$\implies$ Reason is true.

Arithmetic Progressions — Common Difference

- 18. 18.** Assertion (A): The common difference of the Arithmetic Progression $5, 2, -1, -4, \dots$ is -3 . Reason (R): The common difference d of an AP is given by $d = a_n - a_{n-1}$.

Answer: (A)

Solution:

1. Calculate the common difference by subtracting the first term from the second term.
2. Check if the formula $d = a_2 - a_1$ matches the given definition in the Reason.
3. Verify $2 - 5 = -3$.
4. Conclude that both statements are true and the reason explains the assertion.

Derivation: $d = a_2 - a_1 = 2 - 5 = -3$. Since the definition $d = a_n - a_{n-1}$ is the standard formula for common difference, both A and R are true and R explains A.

Arithmetic Progressions — Common Difference

- 19. 19.** Assertion (A): The common difference of the Arithmetic Progression $5, 2, -1, -4, \dots$ is -3 . Reason (R): The common difference d of an Arithmetic Progression $a_1, a_2, a_3, \dots$ is given by $d = a_n - a_{n-1}$ for $n > 1$.

Answer: (A)

Solution:

1. Identify the first two terms of the given AP: $a_1 = 5$ and $a_2 = 2$.
2. Calculate the common difference using the formula $d = a_2 - a_1$.
3. Substitute the values: $d = 2 - 5 = -3$.
4. Confirm that the Assertion is true and the Reason provides the correct formula used to derive it.

Derivation:

$$d = a_2 - a_1 = 2 - 5 = -3$$

Arithmetic Progressions — Common Difference

- 20. 20.** Assertion (A): The common difference of the Arithmetic Progression 5, 2, -1, -4, ... is -3. Reason (R): The common difference d of an Arithmetic Progression $a_1, a_2, a_3, \dots$ is given by $d = a_n - a_{n-1}$.

Answer: (A)

Solution:

1. Calculate the common difference from the sequence: $d = a_2 - a_1 = 2 - 5 = -3$.
2. Verify the definition of common difference provided in the Reason.
3. Since the calculation matches the statement and the definition is the standard mathematical formula, both A and R are true and R explains A.

Derivation:

$$d = a_2 - a_1 = 2 - 5 = -3. \text{ Formula: } d = a_n - a_{n-1} \text{ holds true.}$$

Arithmetic Progressions — Common Difference

- 21. 21.** Assertion (A): The common difference of the Arithmetic Progression 5, 2, -1, -4, ... is -3. Reason (R): The common difference d of an AP is given by $d = a_n - a_{n-1}$.

Answer: (A)

Solution:

1. Calculate the common difference by subtracting the first term from the second term.
2. Check if the formula stated in the reason is the definition of the common difference for an AP.
3. Verify the calculation: $2 - 5 = -3$.

Derivation: $d = a_2 - a_1 = 2 - 5 = -3$. Reason is the standard definition of common difference.

Arithmetic Progressions — Common Difference

- 22. 22.** Assertion (A): The common difference of the Arithmetic Progression 5, 2, -1, -4, ... is -3. Reason (R): The common difference d of an Arithmetic Progression is given by $d = a_n - a_{n-1}$.

Answer: (A)

Solution:

1. Identify the terms of the AP: $a_1 = 5, a_2 = 2$.
2. Calculate the common difference using the formula $d = a_2 - a_1$.
3. Substitute values: $d = 2 - 5 = -3$.
4. Verify the reason: The formula $d = a_n - a_{n-1}$ is the standard definition of common difference in an AP.
5. Since both statements are true and the reason correctly explains the assertion, the answer is (A).

Derivation:

$$d = a_2 - a_1 = 2 - 5 = -3$$

Arithmetic Progressions — Common Difference

- 23. 23.** Assertion (A): The sum of the first n terms of an arithmetic progression is given by $S_n = \frac{n}{2}[2a + (n - 1)d]$, where a is the first term and d is the common difference. Reason (R): The sum of the first n terms of an AP can be calculated as $S_n = \frac{n}{2}(\text{first term} + \text{last term})$.

Answer: (B)

Solution:

1. The formula $S_n = \frac{n}{2}[2a + (n - 1)d]$ is the standard formula for the sum of n terms of an AP, making Assertion (A) true.
2. The formula $S_n = \frac{n}{2}(a + l)$ where l is the last term (a_n) is also a valid and standard formula for the sum of n terms, making Reason (R) true.
3. While both are true, Reason (R) is a different formulaic representation and does not act as the logical explanation for the derivation of the formula provided in the Assertion.

Derivation:

$$S_n = \frac{n}{2}(a + a_n)$$

$$\implies S_n = \frac{n}{2}(a + a + (n - 1)d) = \frac{n}{2}(2a + (n - 1)d)$$

Arithmetic Progressions — sum of n terms of an AP

- 24. 24.** Assertion (A): The common difference of the Arithmetic Progression $5, 2, -1, -4, \dots$ is -3 . Reason (R): The common difference d of an AP $a_1, a_2, a_3, \dots$ is given by $d = a_n - a_{n-1}$.

Answer: (A)

Solution:

1. Check the validity of Assertion (A): Calculate $d = a_2 - a_1 = 2 - 5 = -3$. The assertion is true.
2. Check the validity of Reason (R): The definition of common difference $d = a_n - a_{n-1}$ is the standard formula used in NCERT. The reason is true.
3. Determine if R explains A: Since the common difference formula is exactly what was used to verify the assertion, R is the correct explanation of A.

Derivation:

$$d = a_2 - a_1 = 2 - 5 = -3$$

$$\implies \text{Assertion is true. Formula } d = a_n - a_{n-1} \text{ is correct definition}$$

$$\implies \text{Reason is true.}$$

Arithmetic Progressions — Common Difference

- 25. 25.** Assertion (A): If the n^{th} term of an Arithmetic Progression is given by $a_n = 3 + 4n$, then the common difference of the AP is 4. Reason (R): The common difference d of an AP is given by the difference between consecutive terms $a_n - a_{n-1}$.

Answer: (A)

Solution:

1. Find the first term a_1 by substituting $n = 1$ in $a_n = 3 + 4n$.
2. Find the second term a_2 by substituting $n = 2$ in $a_n = 3 + 4n$.
3. Calculate the common difference $d = a_2 - a_1$.
4. Observe that the coefficient of n in a linear expression of n represents the common difference.

Derivation:

$$a_1 = 3 + 4(1) = 7$$

$$a_2 = 3 + 4(2) = 11$$

$$d = a_2 - a_1 = 11 - 7 = 4$$

Since $d = 4$, Assertion is true.

The formula $d = a_n - a_{n-1}$ is the standard definition, so Reason is true and is the correct explanation.

Arithmetic Progressions — general term of an AP

- 26. 26.** Assertion (A): The sum of the first n terms of an arithmetic progression is given by $S_n = 3n^2 + 5n$. The common difference of this AP is 6. Reason (R): For an AP where $S_n = an^2 + bn$, the common difference is always $2a$.

Answer: (A)

Solution:

1. To find the common difference from the sum of n terms, calculate S_1 and S_2 .
2. $S_1 = a_1 = 3(1)^2 + 5(1) = 8$.
3. $S_2 = a_1 + a_2 = 3(2)^2 + 5(2) = 12 + 10 = 22$.
4. Calculate $a_2 = S_2 - S_1 = 22 - 8 = 14$.
5. Common difference $d = a_2 - a_1 = 14 - 8 = 6$. Assertion is true.
6. For a general sum $S_n = An^2 + Bn$, $a_n = S_n - S_{n-1} = A(n^2 - (n-1)^2) + B(n - (n-1)) = A(2n-1) + B = 2An - A + B$. Since a_n is a linear function of n , the common difference is the coefficient of n , which is $2A$. Reason is true and correctly explains the assertion.

Derivation:

$$S_n = 3n^2 + 5n$$

$$a_n = S_n - S_{n-1}$$

$$a_n = 3n^2 + 5n - [3(n-1)^2 + 5(n-1)]$$

$$a_n = 3n^2 + 5n - [3(n^2 - 2n + 1) + 5n - 5]$$

$$a_n = 3n^2 + 5n - 3n^2 + 6n - 3 - 5n + 5$$

$$a_n = 6n + 2$$

$$d = a_n - a_{n-1} = (6n + 2) - (6(n-1) + 2) = 6$$

- 27. 27.** Assertion (A): The common difference of the Arithmetic Progression $5, 2, -1, -4, \dots$ is -3 . Reason (R): The common difference d of an Arithmetic Progression is calculated by the formula $d = a_n - a_{n-1}$ for any $n > 1$.

Answer: (A)

Solution:

1. Identify the terms $a_1 = 5$ and $a_2 = 2$.
2. Calculate the common difference using $d = a_2 - a_1$.
3. Verify the result $d = 2 - 5 = -3$.
4. Confirm that the formula in Reason (R) is the definition of common difference and correctly explains the calculation in (A).

Derivation:

$$d = a_2 - a_1 = 2 - 5 = -3. \text{ Formula } d = a_n - a_{n-1} \text{ is standard.}$$

- 28. 28.** If $\sin \theta = \frac{3}{5}$, then the value of $\cos \theta$ is:

Answer: (C)

Solution:

1. Use the identity $\sin^2 \theta + \cos^2 \theta = 1$.
2. Substitute the given value: $(\frac{3}{5})^2 + \cos^2 \theta = 1$.
3. Solve for $\cos^2 \theta = 1 - \frac{9}{25} = \frac{16}{25}$.
4. Take the square root to find $\cos \theta$.

Derivation:

$$\cos \theta = \sqrt{1 - \sin^2 \theta} = \sqrt{1 - (\frac{3}{5})^2} = \sqrt{1 - \frac{9}{25}} = \sqrt{\frac{16}{25}} = \frac{4}{5}$$

- 29. 29.** The value of $\tan 30^\circ \cdot \cot 30^\circ$ is:

Answer: (B)

Solution:

1. Recall the trigonometric identity $\tan \theta \cdot \cot \theta = 1$ for any angle θ .
2. Alternatively, substitute $\tan 30^\circ = \frac{1}{\sqrt{3}}$ and $\cot 30^\circ = \sqrt{3}$.
3. Multiply the values to get the result.

Derivation:

$$\tan 30^\circ \cdot \cot 30^\circ = \frac{1}{\sqrt{3}} \times \sqrt{3} = 1$$

30. 30. If $A = 60^\circ$ and $B = 30^\circ$, then the value of $\sin(A + B)$ is:

Answer: (D)

Solution:

1. Calculate the sum of the angles: $A + B = 60^\circ + 30^\circ = 90^\circ$.
2. Evaluate the sine of the resultant angle: $\sin 90^\circ$.
3. Recall the standard trigonometric table value for $\sin 90^\circ$.

Derivation:

$$\sin(60^\circ + 30^\circ) = \sin 90^\circ = 1$$

31. 31. Assertion (A): The value of $\sin \theta$ is always less than 1. Reason (R): The value of $\sin \theta$ can never exceed 1 for any real value of θ .

Answer: (D)

Solution:

1. Evaluate the assertion: The value of $\sin \theta$ can be equal to 1 when $\theta = 90^\circ$. Therefore, the statement 'always less than 1' is false.
2. Evaluate the reason: The range of $\sin \theta$ is $[-1, 1]$, meaning it can reach 1 but never exceed it. The reason statement is true.

Derivation:

$$\text{Assertion: } \sin(90^\circ) = 1, \text{ so } \sin \theta \leq 1. \text{ Reason: } -1 \leq \sin \theta \leq 1.$$

32. 32. Assertion (A): If $\tan \theta = \cot \theta$, then $\theta = 45^\circ$. Reason (R): $\tan \theta$ and $\cot \theta$ are reciprocal to each other.

Answer: (B)

Solution:

1. Evaluate the assertion: $\tan \theta = \cot \theta \implies \tan \theta = \frac{1}{\tan \theta} \implies \tan^2 \theta = 1 \implies \tan \theta = 1$ (for acute θ). Thus, $\theta = 45^\circ$. Assertion is true.
2. Evaluate the reason: By definition, $\cot \theta = \frac{1}{\tan \theta}$. Reason is true.
3. Check if R explains A: While both are true, the assertion is solved by equating the ratios, not simply by the definition of reciprocals alone, though they are related.

Derivation:

$$\begin{aligned}\tan \theta &= \frac{1}{\tan \theta} \\ \implies \tan^2 \theta &= 1 \\ \implies \theta &= 45^\circ\end{aligned}$$

Introduction to Trigonometry — Trigonometric Ratios of Complementary Angles

33. 33. If $3 \cot \theta = 2$, find the value of $\frac{4 \sin \theta - 3 \cos \theta}{2 \sin \theta + 6 \cos \theta}$.

Answer:

$$\frac{1}{3}$$

Solution:

1. Given $3 \cot \theta = 2$, so $\cot \theta = \frac{2}{3} = \frac{\cos \theta}{\sin \theta}$.
2. Divide numerator and denominator of the expression by $\sin \theta$ to get $\frac{4 - 3 \cot \theta}{2 + 6 \cot \theta}$.
3. Substitute $\cot \theta = \frac{2}{3}$ and simplify.

Derivation:

$$\frac{4 \sin \theta - 3 \cos \theta}{2 \sin \theta + 6 \cos \theta} = \frac{4 - 3 \cot \theta}{2 + 6 \cot \theta} = \frac{4 - 3(2/3)}{2 + 6(2/3)} = \frac{4 - 2}{2 + 4} = \frac{2}{6} = \frac{1}{3}$$

Introduction to Trigonometry — Trigonometric Ratios

34. 34. Evaluate $\frac{\cos 45^\circ}{\sec 30^\circ + \operatorname{cosec} 30^\circ}$.

Answer:

$$\frac{3\sqrt{2} - \sqrt{6}}{8}$$

Solution:

1. Substitute the standard values: $\cos 45^\circ = \frac{1}{\sqrt{2}}$, $\sec 30^\circ = \frac{2}{\sqrt{3}}$, and $\operatorname{cosec} 30^\circ = 2$.
2. Simplify the denominator and rationalize the expression.

Derivation:

$$\frac{1/\sqrt{2}}{2/\sqrt{3} + 2} = \frac{1/\sqrt{2}}{(2 + 2\sqrt{3})/\sqrt{3}} = \frac{\sqrt{3}}{\sqrt{2}(2 + 2\sqrt{3})} = \frac{\sqrt{3}}{2\sqrt{2} + 2\sqrt{6}} = \frac{\sqrt{3}(2\sqrt{6} - 2\sqrt{2})}{8 - 24} = \frac{2\sqrt{18} - 2\sqrt{6}}{16} = \frac{6\sqrt{2} - 2\sqrt{6}}{16} = \frac{3\sqrt{2} - \sqrt{6}}{8}$$

Introduction to Trigonometry — Trigonometric Ratios of Specific Angles

35. 35. If $7 \sin^2 \theta + 3 \cos^2 \theta = 4$, show that $\tan \theta = \frac{1}{\sqrt{3}}$.

Answer:

$$\tan \theta = \frac{1}{\sqrt{3}}$$

Solution:

1. Use the identity $\sin^2 \theta + \cos^2 \theta = 1$ to replace $\cos^2 \theta$.
2. Simplify the equation to isolate $\sin^2 \theta$ and $\cos^2 \theta$.
3. Divide by $\cos^2 \theta$ to get $\tan^2 \theta$.
4. Solve for $\tan \theta$.

Derivation:

$$\begin{aligned}
 7 \sin^2 \theta + 3(1 - \sin^2 \theta) &= 4 \\
 7 \sin^2 \theta + 3 - 3 \sin^2 \theta &= 4 \\
 4 \sin^2 \theta &= 1 \\
 \sin^2 \theta &= \frac{1}{4} \\
 \implies \cos^2 \theta &= 1 - \frac{1}{4} = \frac{3}{4} \\
 \tan^2 \theta &= \frac{\sin^2 \theta}{\cos^2 \theta} = \frac{1/4}{3/4} = \frac{1}{3} \\
 \tan \theta &= \frac{1}{\sqrt{3}}
 \end{aligned}$$

Introduction to Trigonometry — Trigonometric Identities

36. 36. Evaluate the following expression: $\frac{\cos 58^\circ}{\sin 32^\circ} + \frac{\sin 22^\circ}{\cos 68^\circ} - \frac{\cos 38^\circ \csc 52^\circ}{\tan 18^\circ \tan 35^\circ \tan 60^\circ \tan 72^\circ \tan 55^\circ}$.

Answer:

$$1 - \frac{1}{\sqrt{3}}$$

Solution:

1. Use complementary angle relations $\cos(90^\circ - \theta) = \sin \theta$ and $\tan(90^\circ - \theta) = \cot \theta$.
2. Simplify the individual terms in the expression.
3. Substitute the value of $\tan 60^\circ = \sqrt{3}$ and simplify the final result.

Derivation:

$$\begin{aligned}
 \frac{\cos(90^\circ - 32^\circ)}{\sin 32^\circ} + \frac{\sin(90^\circ - 68^\circ)}{\cos 68^\circ} - \frac{\cos 38^\circ \cdot (1/\sin 52^\circ)}{\tan 18^\circ \tan 35^\circ \sqrt{3} \cot 18^\circ \cot 35^\circ} \\
 = \frac{\sin 32^\circ}{\sin 32^\circ} + \frac{\cos 68^\circ}{\cos 68^\circ} - \frac{\cos 38^\circ / \cos 38^\circ}{\sqrt{3}} \\
 = 1 + 1 - \frac{1}{\sqrt{3}} = 2 - \frac{1}{\sqrt{3}}
 \end{aligned}$$

Introduction to Trigonometry — Trigonometric Ratios of Complementary Angles

37. 37. Prove the following trigonometric identity: $\frac{\cos \theta - \sin \theta + 1}{\cos \theta + \sin \theta - 1} = \csc \theta + \cot \theta$, using the identity $\csc^2 \theta = 1 + \cot^2 \theta$.

Answer:

$$\csc \theta + \cot \theta$$

Solution:

1. Divide the numerator and the denominator of the Left Hand Side (LHS) by $\sin \theta$.
2. Recall that $\frac{\cos \theta}{\sin \theta} = \cot \theta$, $\frac{\sin \theta}{\sin \theta} = 1$, and $\frac{1}{\sin \theta} = \csc \theta$.
3. Rearrange the terms to group $\csc \theta$ and $\cot \theta$ together.
4. Use the identity $1 = \csc^2 \theta - \cot^2 \theta$ to substitute the '1' in the denominator.
5. Factorize the denominator using the difference of squares identity $a^2 - b^2 = (a - b)(a + b)$.
6. Cancel the common term in the numerator and denominator to arrive at the RHS.

Derivation:

$$\begin{aligned}
 LHS &= \frac{\frac{\cos \theta}{\sin \theta} - \frac{\sin \theta}{\sin \theta} + \frac{1}{\sin \theta}}{\frac{\cos \theta}{\sin \theta} + \frac{\sin \theta}{\sin \theta} - \frac{1}{\sin \theta}} = \frac{\cot \theta - 1 + \csc \theta}{\cot \theta + 1 - \csc \theta} \\
 &= \frac{\cot \theta + \csc \theta - (\csc^2 \theta - \cot^2 \theta)}{\cot \theta - \csc \theta + 1} \\
 &= \frac{(\cot \theta + \csc \theta) - [(\csc \theta - \cot \theta)(\csc \theta + \cot \theta)]}{\cot \theta - \csc \theta + 1} \\
 &= \frac{(\cot \theta + \csc \theta)[1 - (\csc \theta - \cot \theta)]}{\cot \theta - \csc \theta + 1} \\
 &= \frac{(\cot \theta + \csc \theta)[1 - \csc \theta + \cot \theta]}{\cot \theta - \csc \theta + 1} \\
 &= \cot \theta + \csc \theta = RHS
 \end{aligned}$$

Introduction to Trigonometry — Trigonometric Identities

- 38. 38.** A tower stands vertically on the ground. From a point on the ground, which is 15 m away from the foot of the tower, the angle of elevation of the top of the tower is found to be 60° . The height of the tower is:

Answer: (C)

Solution:

1. Let the height of the tower be h .
2. Given distance from foot = 15 m.
3. In $\triangle ABC$, $\tan(60^\circ) = \frac{\text{height}}{\text{base}}$.
4. $\sqrt{3} = \frac{h}{15}$
5. $h = 15\sqrt{3}$

Derivation:

$$\begin{aligned}
 \tan(60^\circ) &= \frac{h}{15} \\
 \implies \sqrt{3} &= \frac{h}{15} \\
 \implies h &= 15\sqrt{3} \text{ m}
 \end{aligned}$$

- 39. 39.** The angle of elevation of the top of a pole from a point on the ground is 30° . If the pole is 10 m high, the distance of the point from the foot of the pole is:

Answer: (B)

Solution:

1. Let the distance be x .
2. Given height = 10 m, angle = 30° .
3. $\tan(30^\circ) = \frac{10}{x}$
4. $\frac{1}{\sqrt{3}} = \frac{10}{x}$
5. $x = 10\sqrt{3}$

Derivation:

$$\begin{aligned}\tan(30^\circ) &= \frac{10}{x} \\ \implies \frac{1}{\sqrt{3}} &= \frac{10}{x} \\ \implies x &= 10\sqrt{3}\end{aligned}$$

- 40. 40.** If the length of a string of a kite flying at a height of 60 m from the ground is 120 m, then the angle of inclination of the string with the ground is:

Answer: (A)

Solution:

1. Let the angle be θ .
2. Height (opposite) = 60 m, Hypotenuse = 120 m.
3. $\sin(\theta) = \frac{60}{120}$
4. $\sin(\theta) = \frac{1}{2}$
5. $\theta = 30^\circ$

Derivation:

$$\begin{aligned}\sin(\theta) &= \frac{60}{120} = \frac{1}{2} \\ \implies \theta &= 30^\circ\end{aligned}$$